A Multi-Modal Framework for Quantifying Narrator Reliability in ʿIlm ar-Rijal

Using Semantic Sentiment Analysis and Graph Neural Networks



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Introduction

Hadith authentication is fundamental to Islamic scholarship.
Traditional *ʿIlm ar-Rijal* (Hadith narrator biographical studies) methodology is labour-intensive and increasingly inadequate for the digital era. Over 500,000 narrators have been tracked historically through manual processes. Digital collections are growing too large for unaided human review — a scalability crisis

There is a critical need for an automated, scalable solution that preserves classical methodology while leveraging modern computational approaches.

Problem Background

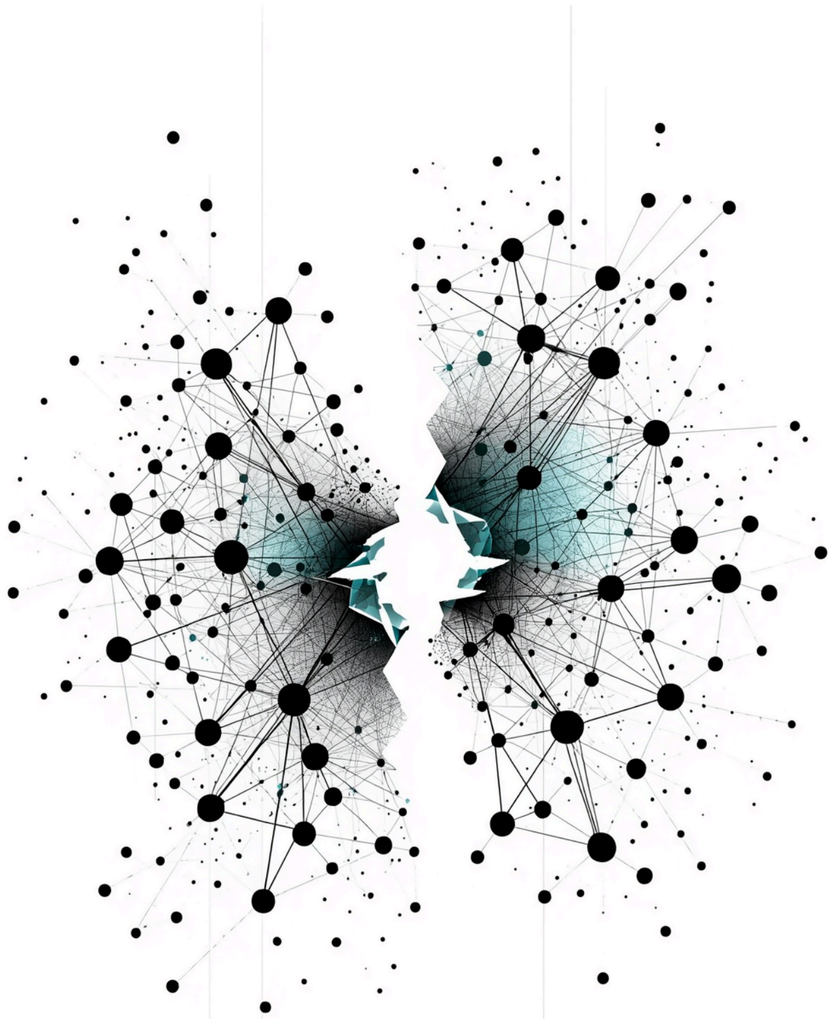
Hadith comprises two components: *Sanad* (chain of narrators) and *Matn* (textual content). Authentication requires assessing each narrator's moral integrity (*ʿadāla)* and memory precision (*ḍabṭ*) through manual cross-referencing of vast biographical texts such as Taqrīb al-Tahḏīb of Ibn Ḥajar al-ʿAsqalāni and the Tahḏīb al-Kamāl of al-Mizzī.

وَحَدَّثَنِي عَنْ مَالِكٍ، عَنِ ابْنِ شِهَابٍ، عَنْ عَلِيِّ بْنِ حُسَيْنِ بْنِ عَلِيِّ بْنِ أَبِي طَالِبٍ، أَنَّ رَسُولَ اللَّهِ صلى الله عليه وسلم قَالَ " مِنْ حُسْنِ إِسْلاَمِ الْمَرْءِ تَرْكُهُ مَا لاَ يَعْنِيهِ " .

And it was related to me (Yahya ibn Yahya al-Laythi) from Mālik, from Ibn Shihāb, from Alī ibn Husayn ibn Alī ibn Abi Ṭālib, that the Messenger of Allah ﷺ said, "Part of the excellence of a man's Islam is that he leaves that does not concern him" (Muwaṭṭa' Mālik, Book 47, Hadith 3). This Hadith is graded as *Ṣaḥīḥ*.

Key challenges include Arabic orthographic ambiguity, name variations across sources, and temporal/geographical verification complexities.

Critical gap: **No end-to-end automated reliability scoring system exists that preserves classical methodology while enabling digital-scale analysis.**



Problem Statement

Current NER models extract names but cannot judge narrator reliability standing.

No existing framework maps **Jarḥ wa Taʿdīl terminology → standardised reliability index**

Transformer models extract chains as graphs (Mosa, 2025; Oumer, 2025)

- But stop at graph construction — no reliability scores assigned
- No use of actual Jarḥ wa Taʿdīl terms to judge trustworthiness
- This project adds a quantitative reliability index + GNN structural analysis

Proposed Solution

End-to-End Pipeline

First complete automated framework: **extraction → disambiguation → scoring → grading**. No existing solution provides this full pipeline from raw biographical texts to final Hadith reliability classification.

Multi-Modal Integration

Combines NER + Knowledge Graph + Graph Neural Networks + Semantic Sentiment Analysis. Probabilistic trust scores (0.0-1.0) per narrator with edge modifiers for transmission risks including ambiguous attribution (*tadlīs*), memory lapses, and geographical impossibilities.

Explainable & Validated

Evaluation against *Kutub as-Sitta* (the Six Canonical Hadith Collections) ground-truth collections.

Research Objectives

Objective 1: Semantic Sentiment Integration

To develop a Transformer-based NLP classification model utilising semantic sentiment analysis to map qualitative scholarly judgments from historical biographies to a standardised, quantitative reliability index bounded between 0.0 and 1.0.

Objective 2: GNN & Knowledge Graph

To construct a multi-relational Knowledge Graph of narrator transmission networks and leverage Graph Neural Networks (GNNs) to evaluate structural topology alongside temporal and geographical features as contributing factors to chain plausibility.

Objective 3: End-to-end Evaluation

To evaluate the multi-modal framework's verification accuracy by benchmarking automated hybrid reliability scores against canonical, ground-truth chain-level Hadith gradings (Ṣaḥīḥ, Ḥasan, Ḍaʿīf).

Problem Scope

Data Sources

Taqrīb at-Tahdhīb (Shamela)

Tahdhīb al-Kamāl (Shamela)

Sanadset 650k

Kutub as-Sitta from HuggingFace hadith_datasets

Ṣaḥīḥ al-Bukhārī

Ṣaḥīḥ Muslim

Sunan Abī Dāwūd

Jāmiʿ at-Tirmiḏī

Sunan an-Nasāʾī

Sunan Ibn Mājah

Technical Scope

Arabic NER, entity disambiguation, reliability scoring, geospatial mapping.

Targeted Scale

- ~1,500 narrators
- ~1,000 *isnād* chains

Out of Scope

- *Matn* criticism (content analysis of Hadith text)
- *Shīʿa* methodological traditions
- Real-time production deployment

Research Gaps from Literature Review

The Disciplinary Split

- Prior text NLP models extract narrator entities but completely discard graph-structural properties (Alghamdi et al., 2025). Conversely, graph architectures evaluate network topology but strip away the rich semantic qualitative data contained within historical text biographies (Mosa, 2025; Mahmoud et al., 2024).

Missing End-to-End Pipelines

- No published study has delivered an integrated, end-to-end framework that unifies entity parsing, Knowledge Graph identity linking, and automated probabilistic reliability scoring into a single, comprehensive computational artifact (Ahmad et al., 2025).

Evaluation Standardisation Deficit

- The domain lacks a unified evaluation baseline or shared benchmark protocol to rigorously test automated chain grading accuracy against the explicit, ground-truth historical classifications compiled across the Six Canonical Books (Kutub as-Sitta) (Farooqi et al., 2024; Mahmoud et al., 2022).

Previous Researches from Literature Review

Author(s) & Year	Topic / Focus Area	Methodology / Tools	Key Contributions & Limitations
Mosa (2025)	Narrator Disambiguation / KG-Transformer	Pretrained Language Models (PLMs), Knowledge Graph-Transformer	Synergized structure and semantics for narrator node embeddings; achieved ~97% accuracy on AR-Sanad 280K-v2. Did not assign reliability scores.
Muther et al. (2023)	Named Entity Recognition (NER)	Transformer-based Arabic NER	Precise extraction and disambiguation of narrator identities from classical texts.
Mahmoud et al. (2024)	Structural Integrity & Networks	Graph Neural Networks (GNN), NarratorsKG, Graph Query Embeddings	Introduced narrator identification utilizing graph query embeddings and a knowledge graph. Did not incorporate semantic richness of biographical texts.
Ghriss (2025)	Reliability Scoring / Sentiment Quantisation	Semantic Sentiment Quantisation, Iterative Co-Update Algorithm (ICUA)	Proposed a graph-based iterative framework in spired by traditional Islamic Hadith scholarship to compute chain-level scores. Did not apply yet, still a sort of feasibility research.
Mahmoud et al. (2022)	Narrator Disambiguation	AraBERT, AR-Sanad 280K dataset	Established a new baseline for narrator disambiguation using large-scale datasets and deep bidirectional representations.
Oumer (2025)	Graph Representation / Isnād Modelling	AraT5, CAMELBERT-CA	Turned Hadith text into relational tuples and graph representations. Explicitly left out narrator reliability scoring and entity linking.
Alghamdi et al. (2025)	Entity Extraction & Fraud Detection	Transformer-based models (AraBERT, MARBERT)	Demonstrated PLMs significantly outperform previous iterations in identifying fraudulent content and extracting technical entities. Did not feed outputs into KG/GNN.
Mghari et al. (2024)	Semantic Sentiment Quantisation	Narrator2Vec, Word Embeddings	Used word embeddings to create dense representations of a narrator's standing, mapping technical terms to reliability weights.
Ahmad et al. (2025)	Isnād Analysis & Reliability	Machine Learning	Confirmed technical viability of automating isnād analysis but noted a critical lack of frameworks capable of quantifying narrator reliability.
Farooqi et al. (2024)	Isnād Modelling	Structured transmission datasets	Contributed foundational work on structured transmission datasets, but highlighted lack of shared benchmark protocols for end-to-end evaluation.
Najeeb (2022)	Isnād Modelling	Sequence-based isnād modelling	Contributed to resolving identities across centuries, though focused primarily on sequence-based extraction.